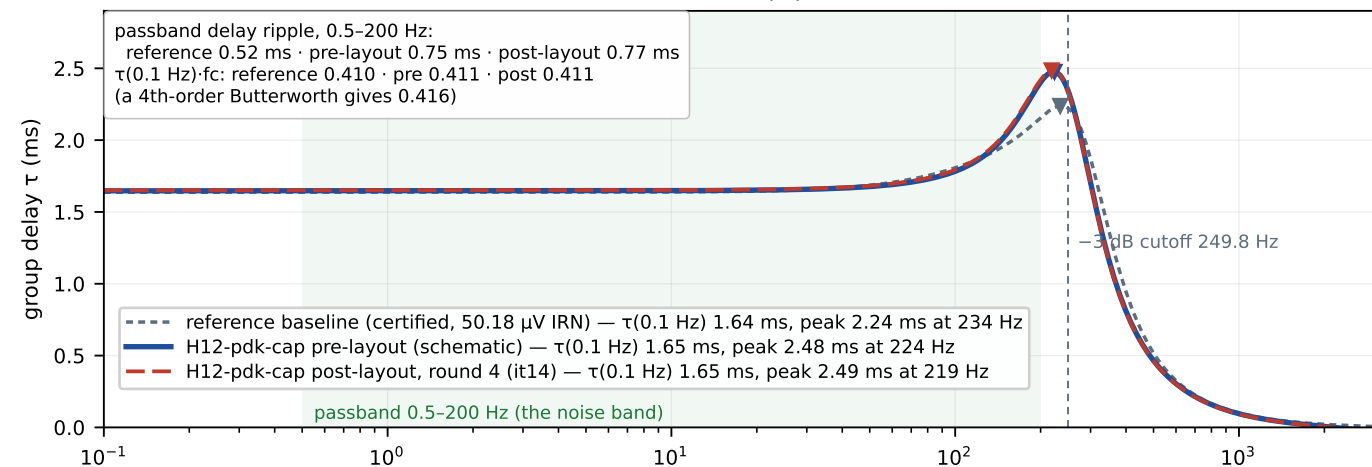
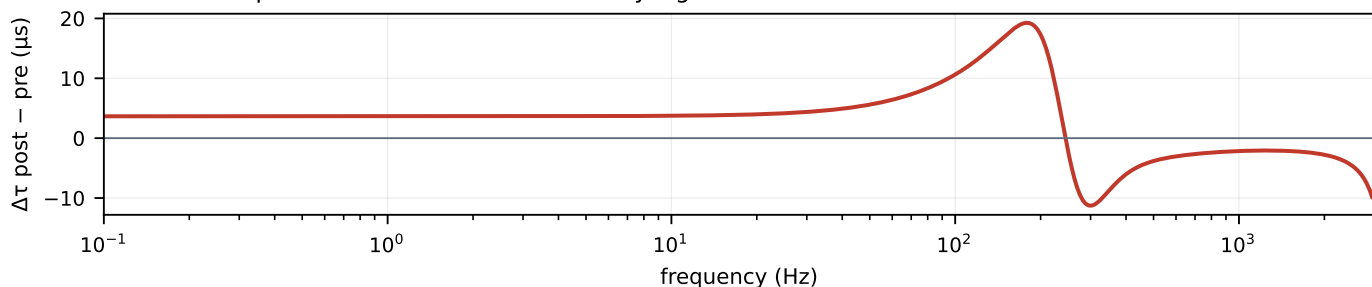


Group delay $\tau(f) = -d\phi/d\omega$ of the 250 Hz low-pass filter (cell `lpf_core`, differential output)

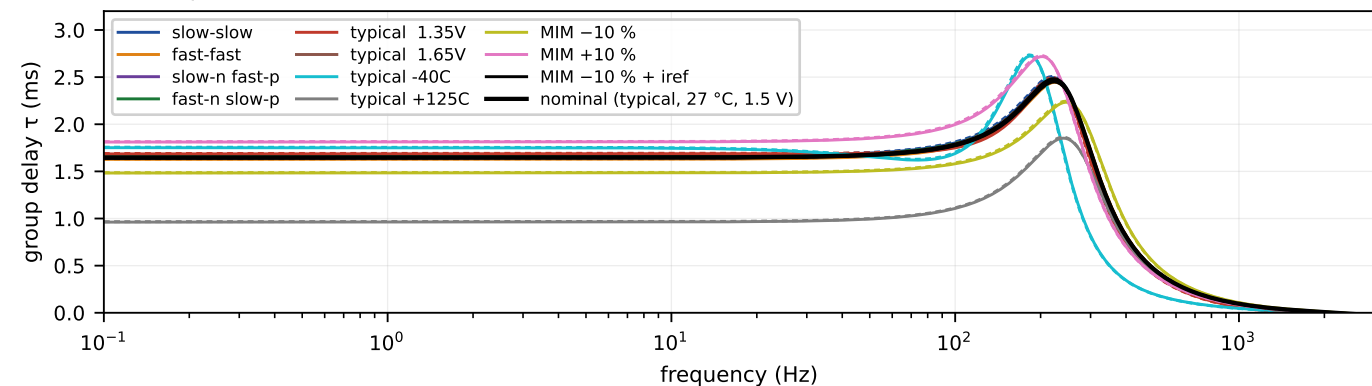
(a) typical process, 27 °C, 1.5 V — flat below ≈ 30 Hz, peaks just below the cutoff, then falls through the stopband (curves stop where $|H|$ reaches the -100 dB floor)



(b) post – pre-layout: parasitics add +3.7 μs at 0.1 Hz (+0.22 %) and +10.0 μs at the peak (+0.40 %); the S-shape around the cutoff is the delay signature of the 1.06 Hz cutoff shift



(c) one-axis corners — solid: pre-layout, dashed: post-layout it14 (same colour); temperature corners use the $\alpha = 1.1$ bias law



(d) mismatch Monte Carlo, 100 paired seeds, typical process, 27 °C, 1.5 V — τ at 0.1 Hz (solid) and at the passband peak (dashed)

